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Education

VIT Bhopal University

Bachelor of Technology in Computer Science and Engineering, AI/ML

2023 – 2027

CGPA: **8.52**

Experience

Infosys Springboard

Nov 2025 – Jan 2026

Machine Learning Intern [Live Demo](#)

- Spearheaded a **team of 5** to build **CliniScan**, an AI-based lung abnormality detection system deployed for radiologists and clinicians.
- Constructed a multi-label chest X-ray diagnostic workflow using **EfficientNet-B0 CNN** with BCEWithLogitsLoss and Adam optimizer on **18,000+ VinBigData DICOM images** annotated by certified radiologists.
- Delivered clinical benchmarks: **Macro F1 = 0.85, Micro F1 = 0.89, Sensitivity = 0.88, Specificity = 0.88, Macro AUC-ROC = 0.92.**
- Executed lesion localization via **YOLOv8 (Ultralytics)** at **1024×1024** resolution, attaining **mAP@0.5 = 0.84** and **mAP@0.5:0.95 = 0.78.**
- Embedded **Grad-CAM visualization** and launched a live inference endpoint on **Hugging Face Spaces** targeting clinical end-users.

IBM India Pvt. Ltd.

May 2025 – Jul 2025

Summer Industrial Internship – Generative AI [GitHub](#)

- Architected enterprise-grade Generative AI workflows on **IBM Watsonx.ai** and IBM Cloud, boosting model output reliability by **27%.**
- Refined RAG workflows by integrating LLMs with improved embedding selection and re-ranking, raising document fetch precision by **18%.**
- Crafted custom prompt templates and tuned context windows, slashing token usage by **22%** and average latency by **15%.**
- Shipped **MediBot**, an AI-powered medical triage chatbot via transformer-based classification and RAG, achieving **92%+ routing precision** and F1 score of **0.91.**

Projects

SmartOps — Agentic AI Business Automation Platform [GitHub](#) [Live Demo](#)

Mar 2026

- Developed an **agentic AI** platform automating **5+ business workflows** (logistics, inventory, forecasting, routing, decisions) using **multi-agent architecture** and **LLM-based reasoning**, cutting manual operational overhead by **40%.**
- Designed **4 specialized autonomous agents** (Inventory, Logistics, Decision, Assistant) with a modular analytics engine achieving **87%+ demand forecast precision** across **3 inventory segments.**
- Deployed multi-modal route planning across transport modes, trimming estimated delivery costs by **30%** and carbon emissions by **25%** through market-aware stock suggestions.

AI Lawyer — RAG Legal Assistant [GitHub](#) [Live Demo](#)

Jul 2025

- Assembled a legal assistant leveraging **LangChain**, **FAISS** vector store, and custom embeddings on Hugging Face LLMs, hitting **85%+ clause retrieval score** via dense vector search.
- Orchestrated a memory-augmented retrieval-generation workflow, lowering hallucination rate in context-aware legal interactions by **20%.**

PIPO – AI-Based Voice Assistant [GitHub](#)

Feb 2025 – Apr 2025

- Built a multilingual voice assistant for real-time task automation with **ASR** and **TTS** modules, attaining **1.2s** response latency and **92% command recognition accuracy.**
- Fused a multimodal NLP stack (**BERT, SpaCy, GPT-2**) for intent inference across **10+ command categories** supporting system-level operations.
- *Publication:* M. R. Bhagat et al., “PIPO: AI-Based Voice Assistant,” **IEEE Global AI Summit 2025**, Noida, pp. 969–974. doi: [10.1109/AISummit66170.2025.11411619](https://doi.org/10.1109/AISummit66170.2025.11411619)

Technical Skills

Languages: Python, C++, SQL, HTML, CSS **Cloud/MLOps:** AWS, IBM Cloud, Microsoft Azure, Docker, CI/CD

ML/DL Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, Neural Networks, EfficientNet, YOLO

LLM/NLP: LangChain, RAG, FAISS, Hugging Face Transformers, Prompt Engineering, NLTK, ML Models

Tools: Git, Streamlit, OpenCV, Pandas, NumPy, Grad-CAM, pyttsx3, SpeechRecognition

Achievements & Certifications

Economic Times GenAI Hackathon — Finalist

Top 40 teams among **54,000+** participants

Conceptualised *AI Money Mentor*, a Generative AI-powered personal finance assistant for intelligent money management.

Smart India Hackathon (SIH) 2025 — Internal Round Finalist

Top 45 among **950+** teams

Led a team of 6; Problem SIH25045 — *CropSure (Agritech-based Yield Monitoring Solution)*

[IBM Cloud Certification](#) [Microsoft Azure DP-900](#) [Applied Machine Learning](#)

Toastmasters International District 98 — Core Member